☐ CTL Provisional DECISION SHEET (PDSH)
☑ CTL DECISION SHEET (DSH)

Standard(s) (incl. year)	Subclause(s)	Tracking No.
IEC 62930:2017	5.4 Table 3- 12 & 13	DSH 2226
Category		
CABL		
Subject	**Keywords**	**Developed by**
Vertical flame propagation and smoke emission tests for muti-core cables	-muti-core cables - vertical flame propagation - smoke emission	ETF06

Question

According to clause 5.4 in IEC 62930:2017, multi-core constructions are allowed provided that all cores are individually sheathed and comply with all requirements set in IEC 62930:2017.

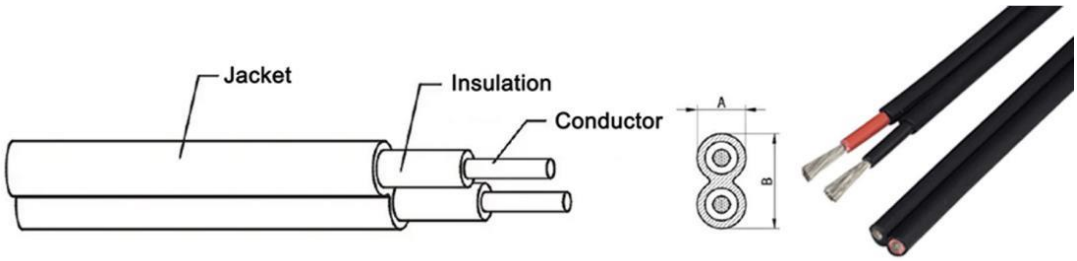


Fig. 1 Typical image and structure model of double-core PV cable

However, when it comes to the vertical flame propagation test and smoke emission test, there seem to be two options for the multi-core cable, 1) separate the multi-core cable into independent single-core cables, and then perform the tests on those single-core cables, respectively; 2) perform the tests directly on the multi-core cable.

Thus, how to perform the vertical flame propagation test and smoke emission test of multi-core cable?

Decision

For multi-core cable, the vertical flame propagation test and smoke emission test shall be carried out directly on the completed multi-core cable, instead of separated cores, so as to keep in line with their actual application scenario.

Explanatory notes

Enquiry:
Date: 2023-06-24

PDSH:
Year of publication: 2023
To be approved at CTL plenary: 2024

DSH
Publication date:2024-09-16

Background:

As multi-core PV cable, e.g. double-core cable, becomes widely used in PV systems, the vertical flame propagation test and smoke emission test of multi-core cable need to be more clear and definite, although it's not detailed in IEC 62930:2017.

Disclaimer: This document is controlled and has been released electronically.
Only the version on the IECEE Website is the current document version.

Rationale:

1 If multi-core cables are separated into single-core cables for testing, the purpose of these tests may not be in line with their actual application scenario, where they are installed as one whole, without being separated.

2 The test results between the multi-core cable and separated single-core cable could vary widely in the smoke emission test. The number of test pieces for multi-core cable would be the same as single-core cable according to IEC61034-2, but a piece of multi-core cable possesses nearly twice or even three times the non-metallic materials of separated single-core cable, thus generating highly different result from single-core cable. The same situation also happens in the vertical flame propagation test of multiple-core cables.